

LAB 1

REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #15

PHARMACY EXPIRY & RE-ORDER DISPATCH ENGINE

Domain: Healthcare & Telemedicine

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Actors:

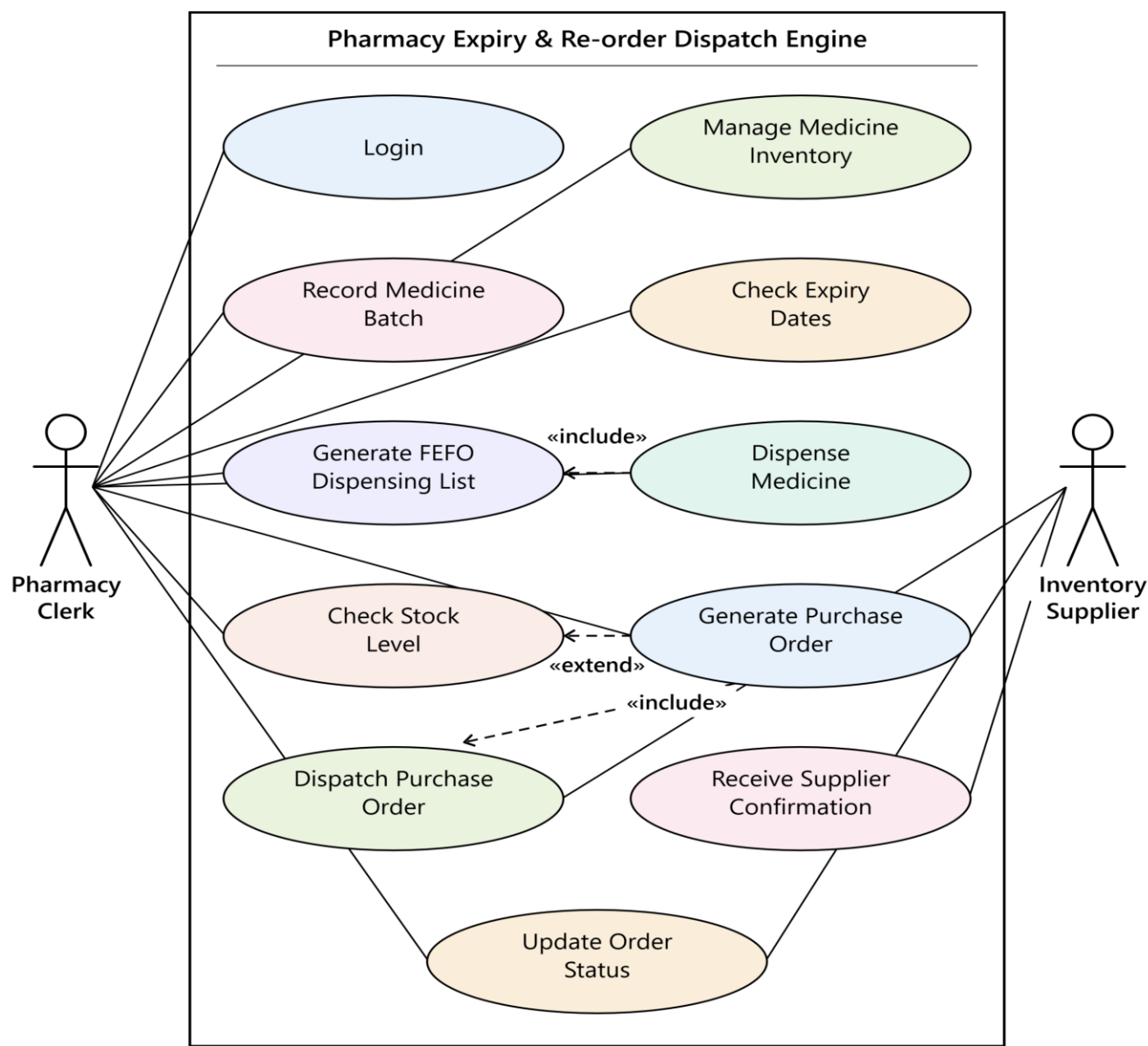
Pharmacy Clerk

Inventory Supplier

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall track medicine batches and their expiry dates.	High	Pass: The system correctly records and displays the batch number and expiry date of each medicine. Fail: Incorrect or missing expiry information is displayed.	To prevent expired medicines from being dispensed and to maintain accurate batch information.
FR-002	Functional	The system shall generate a First-Expired-First-Out (FEFO) dispensing list by prioritizing medicine batches with the nearest valid expiry date.	High	Pass: The batch with the nearest valid expiry date is selected first when multiple valid batches exist. Fail: A newer batch is selected while an older unexpired batch exists.	To ensure medicines are dispensed according to FEFO and reduce medicine wastage due to expiry.
FR-003	Functional	The system shall monitor medicine stock levels against predefined safety thresholds.	High	Pass: The system identifies medicines whose available stock is below the configured safety threshold. Fail: A medicine below the threshold is not identified.	To prevent medicine stock-outs and maintain sufficient inventory.

FR-004	Functional	The system shall automatically generate a purchase order when medicine stock falls below the safety threshold.	High	Pass: A purchase order is generated for every medicine requiring re-order. Fail: No purchase order is generated when stock is below the threshold.	To automate the re-order process and ensure timely replenishment of medicines.
FR-005	Functional	The system shall dispatch the generated purchase order to the Inventory Supplier and record the order status.	High	Pass: The purchase order is successfully dispatched and its status is recorded. Fail: The order is not dispatched or its status is not updated.	To ensure the Inventory Supplier receives the re-order request and the pharmacy can track the order.
NFR-001	Performance & Security	The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below the safety threshold.	High	Pass: The purchase order is generated and dispatched within the defined response target under simulated peak load. Fail: The system fails to generate or dispatch the order within the required response target.	To ensure timely re-ordering while maintaining reliable system operation and security.
NFR-002	Reliability	The system shall maintain accurate medicine, batch, expiry, stock, and purchase-order information during normal operation.	High	Pass: Inventory and purchase-order information remains accurate after system operations. Fail: Stored information is lost, duplicated, or incorrectly updated.	To ensure reliable pharmacy inventory and order management.

2. UML USE-CASE DIAGRAM



USE-CASE FLOW SPECIFICATION

Pharmacy Expiry & Re-order Dispatch Engine

Use Case: Generate Purchase Order

Primary Actor: Pharmacy Clerk

Supporting Actor: Inventory Supplier

1. Preconditions

1. The Pharmacy Clerk is logged into the system.
2. Medicine inventory information is available in the system.
3. Safety-stock thresholds are configured for the medicines.
4. Current medicine stock quantities are available.

2. Postconditions

1. A purchase order is generated for medicines requiring re-order.
2. The purchase order is dispatched to the Inventory Supplier.
3. The purchase order status is recorded in the system.

3. Main Success Scenario

1. The Pharmacy Clerk logs into the system.
2. The system displays the current medicine inventory.
3. The system checks medicine stock levels against their safety thresholds.
4. The system identifies medicines whose stock is below the safety threshold.
5. The system determines the required re-order quantity.
6. The system generates a purchase order for the required medicines.
7. The Pharmacy Clerk reviews the generated purchase order.
8. The Pharmacy Clerk confirms the purchase order.
9. The system dispatches the purchase order to the Inventory Supplier.
10. The system records the purchase order and updates its status.

4. Alternate Flow

A1. No Medicine Requires Re-order

1. The system checks all medicine stock levels.
2. No medicine is found below its safety threshold.
3. The system displays "**No Re-order Required.**"
4. The system does not generate a purchase order.
5. The Pharmacy Clerk can continue managing the inventory.

5. Expected Outcome

The system successfully generates and dispatches a purchase order when medicine stock falls below the configured safety threshold, allowing the pharmacy to maintain adequate inventory.